

ChainCred — Reporter's Guide & Demo Script

This document is for the teammates presenting ChainCred. Read it end to end before the demo day. It covers what the app is, how to talk about the problem, what accounts to prepare, and the exact steps to walk through during the live demo.

Part 1 — Understanding the App (Read This First)

What ChainCred Is

ChainCred is a platform where individuals can collect their academic and professional credentials in one place, have those credentials verified by the organizations that issued them, and share them with potential employers through a simple link. Every verification is recorded as a transaction on the Cardano blockchain, so the record is public, permanent, and tamper-proof.

There are two types of accounts:

Individual (Holder)

- Registers and logs in through a Cardano wallet — no password
- Submits credentials they own (diplomas, certificates, etc.) to a registered organization for verification
- Receives credentials issued directly by organizations straight into their vault
- Shares a portfolio link with specific recipients (e.g. an HR recruiter) — the recipient does not need a ChainCred account or a Cardano wallet to view the portfolio
- Can edit credential details as long as the credential is still pending (not yet verified)
- Can delete individual credentials from their vault (requires wallet confirmation)
- Can log out or delete their account from Settings

Organization

- Also registers and logs in through a Cardano wallet
- Has a dashboard showing all credentials they have issued or verified
- Can issue credentials directly to a registered holder — the credential goes straight into that holder's vault as verified with a real on-chain transaction
- Has a Pending Queue where they see credentials submitted to them for verification — they can approve (which triggers a Cardano transaction and marks it verified) or reject, individually or in bulk
- Can revoke a verified credential — for example, if the organization made an error or the holder's standing changed

What Happens On-Chain

Every meaningful action produces a Cardano transaction on the Preview Testnet:

Action	What Goes On-Chain
Organization approves a credential	Metadata (label 674): credential name, organization, issue date, SHA-256 hash of the document, owner wallet, verified-by organization
Organization issues a credential directly	Same metadata plus the IPFS CID of the uploaded file

All transactions can be seen live on preview.cardanoscan.io.

IPFS and Blockfrost

When a credential has a supporting document (PDF, PNG, JPG, WebP), ChainCred:

1. Computes a SHA-256 hash of the file in the browser using the Web Crypto API
2. Uploads the file to IPFS through the Blockfrost IPFS API
3. Pins the file on IPFS so it persists
4. Stores the IPFS CID and gateway URL alongside the credential

The document is retrievable at <https://ipfs.io/ipfs/<CID>> from anywhere in the world, permanently, independent of ChainCred's servers.

On document privacy: Because the IPFS CID is embedded in the on-chain transaction metadata, anyone who finds the CID can retrieve the underlying document from any public IPFS gateway. This is a known tradeoff — the flip side is that public verifiability of the document is also a feature, since an employer can confirm the file they are looking at matches the on-chain hash exactly. A future version would encrypt documents before uploading so only authorized parties can decrypt them, while keeping the hash publicly verifiable. If this comes up during the demo, frame it as a deliberate tradeoff between verifiability and privacy that a production version would address with encryption.

Part 2 — The Problem Statement

The Core Argument

LinkedIn and similar platforms let you list certifications and upload certificate images, but there is no built-in mechanism for anyone to verify those claims. A person can upload a fake PDF, type in a degree they never earned, or claim dates that are incorrect. Employers who want to verify

have to contact the organization directly — by email or phone — which is slow, inconsistent, and often does not happen at all.

ChainCred's answer is: put the verification record somewhere no one can edit it. Once an organization clicks "Approve" in ChainCred, a Cardano transaction is submitted that permanently records the fact that they certified this credential for this person on this date. That record exists independent of ChainCred. Even if our app goes offline tomorrow, the transaction is still on the blockchain.

How to Frame It

Keep it simple. The one-sentence version: "You can upload a fake certificate to LinkedIn. You cannot fake a blockchain transaction from the organization that issued it."

The three things ChainCred solves:

1. **Authenticity** — credentials have an on-chain verification record, not just an image upload
 2. **Aggregation** — all your credentials are in one place instead of scattered across email attachments and Google Drive
 3. **Easy sharing** — one link, no documents to email back and forth
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Part 3 — Potential Counterarguments

Your panel or audience may raise these. Being ready shows you thought critically about your own work.

"Aren't there already apps that do this? What about Blockcerts or Credly?"

Blockcerts is an open standard for blockchain-anchored certificates, and Credly is a digital badge platform used by major certification bodies. These exist. The honest answer is: ChainCred is an academic proof-of-concept that implements the same core idea — on-chain credential verification — as a working DApp on Cardano. The point is not that no one has thought of this, but that we built it. If the panel pushes on this, acknowledge those platforms and point out the difference: ChainCred uses Cardano specifically (not Ethereum or a permissioned chain), and it is designed for the Philippine academic context where those enterprise platforms are not widely adopted by local organizations.

"What if the organization that issued the credential does not have a ChainCred account?"

This is the biggest practical limitation and you should be upfront about it. ChainCred requires the verifying organization to be a registered user. If CIT-U or TESDA does not have an account,

they cannot approve credentials through the platform. In a real-world deployment, this would require organizational onboarding — the same challenge every B2B platform faces. A future version could use a smart contract to allow any organization to verify using their public key without registering.

"Why would a company trust this over just calling the school?"

Blockchain records are auditable and cannot be changed after the fact. Calling the school is also auditable, but it depends on someone answering the phone and the registrar's records being accurate at the time of the call. The blockchain record is timestamped, publicly visible, and does not require anyone to be available. For high-volume employers hiring many candidates at once, automated verification through something like ChainCred is significantly faster than phone verification. The trust argument is the same as trusting any certificate authority — at some point you trust the organization's wallet signature the same way you trust their stamped paper.

"What if an organization's wallet is hacked?"

The same risk exists for organizational email accounts and physical stamps. A compromised wallet is a key management problem, not a flaw specific to blockchain. In a production system, multi-signature wallets or hardware wallets would be required for organization accounts. This is a known mitigation, the same way email security is handled by requiring MFA.

"Blockchain is public — doesn't that expose private credential data?"

The metadata stored on-chain includes the credential name, organization, issue date, and a SHA-256 hash of the document. The actual document lives on IPFS. There is a tradeoff here: IPFS content is publicly accessible by CID, so anyone who finds the CID can retrieve the file. This is intentional for verifiability — an employer can confirm the document matches the on-chain hash — but it means the document is not strictly private. A production version would encrypt documents on upload so only authorized parties can decrypt them, while the hash remains publicly verifiable. The wallet address is pseudonymous, not directly tied to a real name unless you already know the person's wallet.

"What happens when the testnet resets or Blockfrost goes down?"

The app currently runs on the Cardano Preview Testnet for demo purposes. A production deployment would use mainnet, which has over five years of uninterrupted history. Blockfrost is a managed API service with documented SLAs. If Blockfrost is unavailable, IPFS content is still

accessible through any public IPFS gateway because the content is content-addressed, not Blockfrost-specific.

"Why is there no backend? Doesn't that limit what the app can do?"

ChainCred is intentionally designed as a fully client-side application. The reasoning is that a backend server creates a central point of failure and a central point of trust — if the server goes down, or if the company running it disappears, user data could be lost or inaccessible. By keeping everything in the browser and on decentralized infrastructure (Cardano for records, IPFS for files), the app avoids that dependency entirely. The tradeoff is that cross-browser state — like share link visibility for recipients on different devices — requires a shared backend to work universally, and that is something a production version would address.

Internal note: Do not bring this up proactively. If the panel does not ask about the backend, do not mention it. If they do ask, use the answer above.

"Does the share link work if the recipient opens it on a different browser or device?"

Internal note — do not bring this up proactively. Only answer if asked.

No, and this is a known limitation we are aware of. Share link data and credential data are stored in the holder's browser (`localStorage`). A recipient opening the link on a different browser or device has none of that data locally, so the public profile will appear empty. The share link itself is technically valid — the URL resolves and the page loads — but there are no credentials to display because they live in the holder's browser, not on a shared server. The correct fix is a backend database or encoding credentials into the URL at generation time. We identified both approaches but did not have enough time to implement either without risking the stability of the rest of the app. This is the biggest architectural gap in the current version.

"Why not require the recipient to create a ChainCred account so they can see the credentials?"

Internal note — do not bring this up proactively. Only answer if asked.

We considered this. If the recipient were a registered ChainCred user, their credential data would already live in their own vault and cross-browser sharing would not be a problem at all. However, recipients in our intended use case are HR recruiters and employers — requiring them to install a Cardano wallet, fund it with ADA, and register an account just to view a portfolio creates so much friction that it defeats the purpose of having a share link at all. The no-account approach was the right product decision. The implementation gap is on the infrastructure side — it works correctly within the same browser, but needs a backend to work universally.

"What stops a credential holder from creating a fake organization account and approving their own credentials?"

| Internal note — do not bring this up proactively. Only answer if asked.

Nothing in the current version prevents this. A holder could register a second account as an organization — say "Harvard University" — using a different wallet, submit a credential to that fake org, switch to the org account, and approve it themselves. The result is a "Verified" credential with a real on-chain transaction, but the issuing organization is fraudulent.

The honest answer is: ChainCred cannot prevent this right now. The on-chain record is only as trustworthy as the organization that signed it. Anyone can register an organization account with any name — there is no identity verification layer, no check that "Harvard University" in ChainCred is actually Harvard.

The fix in a production version would be an organization onboarding process — requiring orgs to verify their real-world identity through a trusted third party before being allowed to issue credentials, similar to how SSL certificate authorities verify domain ownership before issuing certificates. A smart contract approach could also tie organization authority to a known public key registry maintained by a trusted body.

The framing to use if pushed: "ChainCred solves the problem of tampered records from real institutions — the blockchain guarantees that what was recorded cannot be changed. It does not yet solve the problem of fake institutions registering in the first place. That requires identity verification infrastructure that goes beyond what we built for this project."

"The delete account screen says it submits a blockchain nullification record — does it actually?"

| Internal note — do not bring this up proactively. Only answer if asked, or avoid demoing account deletion entirely.

No. The UI copy in the delete modal mentions a blockchain nullification record, but the actual implementation just calls `localStorage.clear()`. No on-chain transaction is submitted. This is a discrepancy between what the interface says and what the code does — the on-chain nullification was planned but not implemented. If this comes up, be honest: the feature description in the UI was aspirational and the implementation did not catch up. A real version would submit a transaction marking the wallet address as deactivated.

Part 4 — Account Setup (Do This Before the Demo)

You need a minimum of six Cardano wallets, all funded with test ADA on the Preview Testnet. The recommended split is three organization accounts and three individual accounts.

Wallet to use: Lace — this is what the app was built and tested with. Install Lace on each browser you plan to use, create a new wallet per account, and fund each one with test ADA from the Cardano Preview Faucet.

Each wallet only needs about 5 tADA. Each transaction costs roughly 0.17–0.20 tADA, so 5 tADA is more than enough for the demo.

Account 1 — Organization A (Primary Issuer)

- **Account type:** Organization
- **Suggested name:** Cebu Institute of Technology - University
- **Email:** citu@chaincred.app
- **Organization name:** Cebu Institute of Technology - University
- **Purpose:** Issues a credential directly to User A. Approves a pending credential from User B. Demonstrates revocation.

Register this account first. Keep this wallet active on one browser/tab labeled "Organization A."

Account 2 — Organization B

- **Account type:** Organization
- **Suggested name:** Alliance Software Inc.
- **Email:** alliance@chaincred.app
- **Organization name:** Alliance Software Inc.
- **Purpose:** Approves a pending credential from User C. Demonstrates that multiple organizations exist independently.

Use a different browser profile from Organization A. A browser profile is a separate user environment within the same browser — in Chrome, click the profile icon at the top right and create a new profile. Each profile has its own localStorage and session, so accounts do not interfere with each other. The simplest approach for the demo is to use a completely different browser for each account — for example, Organization A on Chrome, Organization B on Firefox, and user accounts on Edge or additional Chrome profiles.

Account 3 — Organization C

- **Account type:** Organization
 - **Suggested name:** Symph Co.
 - **Email:** [symp@chaincred.app](mailto:symph@chaincred.app)
 - **Organization name:** Symph Co.
 - **Purpose:** Optional third organization. Can issue a credential directly to any user account to show parallel issuance flows. Adds visual depth to the dashboard.
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Account 4 — User A (Primary Demo User)

- **Account type:** Individual
 - **Suggested name:** Juan dela Cruz
 - **Email:** juan@demo.com
 - **Purpose:** Main user the audience follows end-to-end — submits a credential, receives a directly issued credential from Organization A, demonstrates the share link.
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Account 5 — User B

- **Account type:** Individual
 - **Suggested name:** Maria Santos
 - **Email:** maria@demo.com
 - **Purpose:** Submits a credential to Organization A for approval. Used to populate Organization A's Pending Queue so it is not empty during the demo.
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Account 6 — User C

- **Account type:** Individual
 - **Suggested name:** Carlo Reyes
 - **Email:** carlo@demo.com
 - **Purpose:** Submits a credential to Organization B. Demonstrates that organization queues are separate — Organization B only sees its own pending credentials, not Organization A's.
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Run through this the night before:

- ☐ All six Lace wallets are created and funded on Preview Testnet
 - ☐ All six accounts are registered on ChainCred
 - ☐ User B already has a credential submitted to Organization A (Pending status) so the queue is populated when you demo it
 - ☐ User C already has a credential submitted to Organization B for the same reason
 - ☐ A sample PDF or image file is ready for the live issuance demo (keep it small, under 5 MB, so the IPFS upload is fast)
 - ☐ `.env` file has valid `VITE_BLOCKFROST_API_KEY`, `VITE_BLOCKFROST_IPFS_KEY`, and `VITE_APP_WALLET_MNEMONIC` — no transactions go through without these
 - ☐ You have done a full dry run — at least one issuance and one approval — on your machine before demo day
 - ☐ Browser layout is planned: know in advance which browser window shows which account
 - ☐ Do NOT demo account deletion — the UI copy says it submits a blockchain record but the code only clears localStorage. Avoid opening this can of worms during the live demo.
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Part 5 — Demo Script

Follow this order. Have one person operating the app and one person narrating — do not try to do both at once.

Segment 1 — Problem & Purpose (1 minute)

Pull up LinkedIn's certifications section on screen if you can.

"Anyone can add a Harvard certificate here. LinkedIn will not check if it is real. Employers who care enough to verify have to email or call the school, wait days for a response, and hope the registrar gets back to them. ChainCred makes that verification instant, public, and permanent by recording it on the Cardano blockchain."

Show the ChainCred login page briefly so the audience is oriented.

Segment 2 — Tech Stack (1 minute)

Speak to this — do not read a slide verbatim:

- **Frontend:** React with TypeScript. No backend server — the application runs entirely in the browser.
 - **Blockchain:** Cardano Preview Testnet. MeshSDK handles building, signing, and submitting transactions.
 - **IPFS:** Supporting documents are uploaded to IPFS through the Blockfrost IPFS API and pinned for permanence. The content identifier is stored in the on-chain metadata, so the document and the verification record are permanently linked.
 - **Wallet authentication:** Users register and log in with a Lace wallet. No passwords. The wallet address is the user's identity.
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Segment 3 — Live Demo (5 minutes)

Step 1 — Register User A (30 seconds)

Show the registration page. Select "Individual." Fill in name and email. Connect the Lace wallet. Submit. Point out that the wallet address is now the unique identifier for this user — no central database involved.

Step 2 — User A submits a credential (1 minute)

Log in as User A. Open the Credential Vault. Click "Add Credential." Fill in the credential name, pick Organization A as the organization, set a date, and attach a PDF. Submit. Walk the audience through the progress steps on screen: hashing → uploading to IPFS → done.

The credential appears in the vault with a "Pending" badge and a SHA-256 hash.

"The hash is a fingerprint of the exact file uploaded. If even one character of the document changes, the hash changes. That is what gets compared on-chain."

Step 3 — Organization A approves the credential (1 minute)

Switch to the Organization A browser. Go to the Pending Queue — User A's credential is there. Click "Approve." Watch "Anchoring to Cardano..." on screen. When done, click "View on Cardanoscan ↗" — show the live transaction in the browser.

Switch back to User A's vault. The credential now shows "Verified" with the real transaction hash.

"This transaction hash is on the Cardano blockchain permanently. Anyone in the world can look it up on Cardanoscan right now, without asking us."

Step 4 — Organization A issues a credential directly (45 seconds)

On Organization A's Dashboard, click "+ Issue Credential." Search for User A by name. Fill in the credential details and attach a file. Submit. Show the IPFS upload and chain submission. Switch

to User A's vault — the credential appears immediately as Verified with a real transaction hash and a working "View" button that opens the document from IPFS.

Step 5 — Organization A revokes a credential (30 seconds)

On Organization A's Dashboard, find one of User A's verified credentials and click Revoke. Confirm. Switch to User A's vault — the credential now shows "Revoked."

Step 6 — User A generates a share link (45 seconds)

On User A's account, go to Share Center. Click "Generate Link." Enter a recipient name such as "Google HR." Confirm. The link is copied to clipboard. Open the copied link in an incognito window on the same browser. The public profile page loads with User A's credentials — verified credentials are clearly labeled with their Cardanoscan links and IPFS document buttons.

"The recipient — in this case an HR recruiter — does not need a ChainCred account or a Cardano wallet. They just open the link."

Step 7 — Credential editing (30 seconds, optional)

Show a credential that is still in Pending status. Click Edit, change the name, save. Then show a Verified credential — the Edit button is gone. Verified credentials are locked.

Segment 4 — Code Highlight (2 minutes)

Highlight 1 — Issuance transaction with IPFS CID in metadata (`IssueCredentialModal.tsx`)

Show the `submitToChain` function. Call out:

- `MeshWallet` initialized with a mnemonic handles signing
- The metadata object goes under label `674` — a Cardano community convention for human-readable metadata
- The `ipfs_cid` field ties the on-chain record directly to the IPFS document
- `trim64` enforces Cardano's per-field 64-byte string limit

Highlight 2 — IPFS upload and pin via Blockfrost (`ipfsStorage.ts`)

Show `uploadToIPFS` and `pinOnIPFS`. Call out:

- The file posts to Blockfrost's `/ipfs/add` endpoint
- The returned `ipfs_hash` is the CID
- A second call to `/ipfs/pin/add/:cid` prevents the file from being garbage-collected
- The public gateway URL is returned and stored alongside the credential so it can be opened later

Segment 5 — Challenges & Reflection (1 minute)

Hardest part to build: The share link system. Share links work correctly within the same browser — the holder generates a link, a recipient opens it in an incognito window on the same machine, and the portfolio loads with full access control. However, because all data lives in `localStorage`, a recipient on a completely different browser or device sees an empty profile. The credential data simply does not exist in their browser. The correct fix is either a backend database or encoding the credential snapshot directly into the URL at generation time. We designed and partially implemented the second approach — HMAC-signed tokens with expiry — but reverted it because it solved the security side without solving the data portability side. We ran out of time to implement it fully.

A decision we would revisit: We designed the share link so that recipients do not need a ChainCred account. This was the right product call — requiring HR recruiters to install a Cardano wallet to view a portfolio is too much friction. But this decision only works cleanly with a backend. Without one, there is no way to serve credential data to a browser that has never seen it before. In hindsight, the no-backend architecture and the no-account-for-recipients decision were in conflict, and we only fully realized that late in development.

What we would improve given more time:

- Add a lightweight backend or serverless function to store credential snapshots so share links work cross-browser
- Encrypt documents before IPFS upload so the file is not publicly accessible by CID alone
- Move transaction signing to a secure backend service so the mnemonic is not in the browser environment
- Add a Plutus or Aiken validator to enforce credential rules at the smart contract level rather than relying on application logic
- Deploy to Cardano mainnet

Part 6 — Edge Cases & System Behavior

This section documents how the system behaves in non-happy-path scenarios. Know these before the demo so you are not caught off guard.

Logging in with a wallet that has no account Login fails silently — the app finds no matching profile for that wallet address and does not proceed.

Submitting a credential without attaching a file Blocked at validation. The submit button triggers an error: "Please fill in all fields and upload a document." A file is required.

Submitting a credential to an organization not in the system The organization name field has a dropdown that only shows registered organizations. If the holder types a name that does not match any registered org, `organizationWallet` will be empty. The credential is added to the holder's vault as Pending, but it will not appear in any organization's Pending Queue because there is no matching wallet to route it to.

Editing a verified credential The Edit button is only shown for credentials in Pending status. Once a credential is verified, the Edit button is completely gone — not just disabled.

Deleting a credential Requires wallet confirmation — the holder must reconnect their Lace wallet to authorize the deletion. The on-chain record is unaffected; only the local vault entry is removed.

What happens to credentials after a holder deletes their account Account deletion calls `localStorage.clear()`, which wipes the holder's entire vault. From the holder's side, everything is gone. However, the organization's dashboard reads credentials from all vault keys in localStorage — since the holder's vault key is now gone, those credentials will also disappear from the org's dashboard on the next refresh. On-chain records are unaffected.

Revoking a credential twice Not possible through the UI. The Revoke button is hidden for credentials that are already in Revoked status.

Opening a share link on a different browser or device The page loads but the public profile appears empty. Credential data lives in the holder's `localStorage` and is not accessible from another browser. This is the known cross-browser limitation described in Segment 5.

Toggling a share link to revoked — does it work cross-browser? No. Revocation state is also stored in `localStorage`. If the holder revokes a link, it blocks access only in the same browser. A recipient on a different device is unaffected.

Part 7 — Quick Reference

Navigation by Account Type

Screen	Who Sees It
Credential Vault	Individual accounts
Share Center	Individual accounts
Public Profile (own)	Individual accounts
Settings	Both account types
Organization Dashboard	Organization accounts
Organization Pending Queue	Organization accounts

Credential Statuses

Status	Meaning
Pending	Submitted by holder, awaiting organization review
Verified	Approved by organization — on-chain transaction exists
Rejected	Organization rejected the submission
Revoked	Organization revoked a previously verified credential

Key URLs

- App: `http://localhost:5173` (local) or your deployed URL
- Cardanoscan Preview: `https://preview.cardanoscan.io/transaction/<txHash>`
- IPFS Gateway: `https://ipfs.io/ipfs/<cid>`
- Preview Faucet: `https://docs.cardano.org/cardano-testnets/tools/faucet/`
- Lace Wallet: `https://www.lace.io/`

Things That Will Break the Demo

- If `VITE_BLOCKFROST_API_KEY` or `VITE_APP_WALLET_MNEMONIC` is missing or wrong, no transactions go through. Test the day before.

- The mnemonic wallet must have enough tADA. Each transaction is roughly 0.17–0.20 tADA. Fund with at least 5 tADA.
- Using the same browser for two accounts means logging into one logs you out of the other. Use separate browser profiles.
- Cardano transactions take a few seconds. Do not click anything while "Anchoring to Cardano..." is on screen.
- The IPFS upload can be slow on a weak connection. Do the dry run on the same network you will use on demo day.
- Do not open the Settings page and click "Delete account" during the demo. The UI mentions a blockchain nullification record that is not actually implemented.
- Do not open the share link in a different browser or on a different device during the demo — the profile will appear empty. Use an incognito window on the same browser instead.